

MID-SEMESTER EXAMINATION, April-2025

Computer Networking: Security (CSE 3752)

Program: B.Tech (CSE/CSIT)
Full Marks: 30

Semester: 6th
Time: 2 Hours

Subject/Course Learning Outcome	*Taxonomy Level	Ques. Nos.	Marks
Understand the various types of security controls and fundamental security concepts.	L1	1	6
Explain the importance of using appropriate cryptographic solutions like PKI, encryption levels, tools, obfuscation, hashing, digital signatures, and certificates.	L3	2,3,4	18
Explain common threat actors, their motivations, and various threat vectors, along with understanding different types of vulnerabilities.	L2	5	6
Analyze the indicators of malicious activity, based on a given scenario and Explain the purpose of mitigation techniques used to secure the enterprise.			
Explain the importance of the security implications of different architecture models, including cloud, serverless, microservices, and network infrastructure			
Apply security principles in different infrastructure scenarios including device placement, security zones, and network appliances			

*Bloom's taxonomy levels: Remembering (L1), Understanding (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

Answer all questions. Each question carries equal mark.

1. (a) Briefly explain various control types with respect to security concern that help organizations to carry out the operations smoothly at their end. 2
- (b) Justify that, the three principles of CIA triad when works in harmony create a robust defense against cyber threats. 2

(c) Briefly highlight the importance of the following concepts associated with the change management process for mitigating the potential disaster effect. 2

- (i) Impact analysis (ii) Test results (iii) Back out plan
(iv) Maintenance window.

2. (a) With the help of suitable diagram differentiate the operation of symmetric and asymmetric encryption technique. 2

(b) What is meant by public and private key associated with public key infrastructure? Distinguish the use of both for the applications (i) Confidentiality (ii) Authentication. 2

(c) Using the keyword "EDUCATION", construct the 5 X 5 matrix key for play fair ciphering and compute the cipher text for the plaintext "HELLO WORLD". Assume the letters I & J share the same cell in key matrix. 2

3. (a) State, how DES algorithm gives a better security as compared to the classical symmetric cryptographic algorithms. 2

(b) Given a 8-bit input as (3A)_{Hex} to the following permutation table. 2

4	7	1	2
6	8	3	5

Find: (i) Respective 8 bit permuted output (in Hex form)

(ii) Inverse permutation table

(c) How many S boxes are used in DES? Given the elements of the 2nd row (i.e. row no. "01") in a S box used in DES as 14, 4, 13, 1, 2, 15, 11, 8, 3, 10, 6, 12, 5, 9, 0 & 7. If the 6 bits input to the S box is "011011". Determine the corresponding 4 bit output? 2

4. (a) Given the plaintext "MEETING POSTPONED" to AES 128 bit algorithm. If the 128 bit key used for encryption as (02020202020202020202020202020202)Hex, then 2

(a) Show the original contents of state array, displayed as a 4 X 4 matrix.

(b) Show the value of state array after initial Add Round Key transformation.

- (b) State the use of RSA algorithm in data communication following cryptography process. Using RSA algorithm, find the value of 'e' if $p=3$, $q=11$ and $d=7$. 2
- (c) A pair of sender and receiver want to use the Diffie-Hellman algorithm to securely share a secret key. They agree to have the prime number and primitive root as 23 and 7 respectively. If the private key used by the sender is 5 and the receiver is 3, then compute : 2
- (i) The data they transmitted to each other for sharing the secret key.
 - (ii) The value of shared key received by each of them. Also, justify that both of them has received the same key.
5. (a) An individual without a lot of experience in IT launches a cyber attack, using readily available tools to disrupt a local government website temporarily. Define threat actor and identify the category in which this scenario best fits. 2
- (b) Employees in a company start using a cloud storage service without authorization, bypassing official IT protocols. Explain the term that best describes this situation. 2
- (c) State the difference between spear phishing from regular phishing. 2

End of Questions

END-SEMESTER EXAMINATION, May-2025

Computer Networking: Security (CSE 3752)

Program: B.Tech (CSE/CSIT)
Full Marks: 60

Semester: 6th
Time: 3 Hours

Subject/Course Learning Outcome	*Taxonomy Level	Ques. Nos.	Marks
Understand the various types of security controls and fundamental security concepts.	L1	1	6
Explain the importance of using appropriate cryptographic solutions like PKI, encryption levels, tools, obfuscation, hashing, digital signatures, and certificates.	L3	2,3,4	18
Explain common threat actors, their motivations, and various threat vectors, along with understanding different types of vulnerabilities.	L2	5,6	12
Analyze the indicators of malicious activity, based on a given scenario and explain the purpose of mitigation techniques used to secure the enterprise.	L2	7,8	12
Explain the importance of the security implications of different architecture models, including cloud, serverless, microservices, and network infrastructure.	L2	9	6
Apply security principles in different infrastructure scenarios including device placement, security zones, and network appliances.	L2	10	6

*Bloom's taxonomy levels: Remembering (L1), Understanding (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

Answer all questions. Each question carries equal mark.

1. (a) List out the common control categories associated with security management of an organization. With the help of suitable examples briefly explain, how technical control improves the security performance. 2

- (b) Discuss the significance of non-repudiation by highlighting its key aspects with respect to cyber communication. 2
- (c) Briefly illustrate the following factors that has a noticeable impact on change management. 2
- (i) Downtime (ii) Service restart (iii) Application restart
(iv) Legacy applications
2. (a) With the help of suitable examples differentiate between substitution and transposition based symmetric cryptography. 2
- (b) State the significance of the two different keys used in public key cryptography. Mention, how these two keys are used for the following two scenarios using public key infrastructure: 2
- (a) Bob wants to send message to Alice securely.
(b) Bob wants to make sure that the message received by him is from Alice only.
- (c) Encrypt the plaintext "EXAMINATION" using the keyword "LEG" with the help of Vigenère cipher. 2
3. (a) State, how both substitution and permutation operations are being a part of DES encryption technique which makes it quite difficult to cryptanalyze. 2
- (b) State the significance of extension permutation table in DES algorithm. Given a 32-bit input as (1234ABCD)_{Hex} to the expansion permutation table used in DES algorithm. Find the respective 48 bit output (in Hexadecimal form). 2
- (c) The 56 bit output of permutation choice 1 table for generating the round keys in DES is (01020304050607)_{Hex}. Find the output (in Hexadecimal form) after circular left shift operation if the same is used to generate the sub key for round 1. 2
4. (a) List out the similarity and differences between AES and DES encryption process. 2
- (b) Given the content of state array after initial round key operation in AES 128 encryption process as (000102030405060708090A0B0C0D0E0F)_{Hex}. 2
- (i) Show the content of output state array (in 4 X 4 matrix form) after Sub byte transformation.
(ii) Show the content of output state array (in 4 X 4 matrix form) after Shift row transformation.
- Assume the elements of 1st row (i.e. row no. '0') in AES S-box as 63, 7C, 77, 7B, F2, 6B, 6F, C5, 30, 01, 67, 2B, FE, D7, A8, 76.

- (c) Joy wants to send the message $M = 13$ to Mike privately, which can only be decrypted by Mike. Which key of Mike, Joy will use to encrypt its message. If Mike's private key is $(33, 7)$ and public key is $(33, 3)$, find the value of encrypted message C. 2
5. (a) A former employee, who is terminated, hacks into a company's database to delete critical customer record to disrupt business operation because of lasting grievance around their termination. Explain the type of motivation that is reflected in the scenario. 2
- (b) List the distinguished feature of a business email compromise attack. 2
- (c) What is meant by vulnerable software and explain how the vulnerabilities occurs. 2
6. (a) Describe a memory injection attack and the risks it poses to application security. 2
- (b) What is meant by Structured Query Language Injection (SQLI) attack? Explain the steps used in this attack to make a system infectious. 2
- (c) List and define the flaws in cryptography process which may lead to a possible cryptographic vulnerability. 2
7. (a) What is meant by DNS cache poisoning associated with DNS attack? Discuss briefly, how DNS sinkhole can be used to prevent DNS attack. 2
- (b) During assessment of security it is noticed that on Monday morning at 9 am, the files of a company's Chief Financial Officer (CFO) are deleted without any warning. The IT Support team restored the data, but on the following Monday morning at 9 am, the files were again deleted. Name and explain the type of attack described by the scenario. 2
- (c) State the essence of indicators of attack as far as the information security in computer network is concerned. Discuss briefly how a blocked content can be used as an indicator of attack. 2
8. (a) In a large enterprise network, the human resources department and the IT department each require isolation from the rest of the company's network. What security technique is adopted to achieve the isolation and why. 2

- (b) Name and define the method that limits the impact of breach. 2
- (c) List the different ways by which the cyber intruders are prevented from moving freely across the entire network. 2
- 9 (a) List and highlight the use of key elements for a multilayer strategy required to be implemented towards achieving effective network security. 2
- (b) Explain briefly the micro-services as a secured architecture listing out the key benefits of it. 2
- (c) With the help of suitable diagram, explain the function of each of the four Supervisory Control and Data Acquisition (SCADA) system levels. 2
- 10 (a) With the help of suitable examples highlight the attributes associated with different categories of devices in the context of infrastructure security. 2
- (b) List and define the various data types considered as a regulated data required to be protected from security breach in organizations. 2
- (c) A company notices that one of its web servers is receiving significantly more traffic than others, leading to performance issues. To overcome the issue the network engineer uses the load balancer to distribute the traffic among the web servers on rotational basis. Name and explain the scheduling technique used by the load balancer that best suites the given scenario. 2